

Chapter 19: Matroids

The concept of matroids can be used to prove results in extremal set theory and graph theory, though they are also of interest in their own right. The following draws on the presentation in B. Bollobás, *Combinatorics*.

19.1 Three Theorems

Here are three famous results that can be proven using matroids. We saw Hall's theorem as a characterization for matching in bipartite graphs. Here is another formulation. Given a collection of sets, a ***system of distinct representatives*** is a distinct element out of each set. An obvious necessary condition is that for any sub-collection of k sets, the union of that sub-collection has at least k elements. Hall showed that this condition is sufficient:

THEOREM. [Hall] *A collection $\mathcal{C}$ of sets has a system of distinct representatives iff for all sub-collections $\mathcal{C}'$ sets it holds that*

$$\left| \bigcup \mathcal{C}' \right| \geq |\mathcal{C}'|.$$

A related problem is where the universe is some vector space and the sets then sets of vectors. The task here is to find for each set a representative such that the group of representatives is linearly independent. Again there is an obvious necessary condition: for any sub-collection of k sets, the rank of the union must be at least k . This condition is sufficient:

THEOREM. [Rado] *A collection $\mathcal{C}$ of sets of vectors has a system of distinct independent representatives iff for all sub-collections $\mathcal{C}'$ it holds that*

$$\text{rank} \left(\bigcup \mathcal{C}' \right) \geq |\mathcal{C}'|.$$

For a different type of result, consider a graph and a palette of k colors. The task is to color each edge such that the resultant coloring does not have a monochromatic cycle. Again there is an obvious necessary condition. The maximum number of edges that can be colored with one particular color is $n - 1$, where n is the order. And indeed this holds true for all subgraphs H . Surprisingly this is sufficient:

THEOREM. [Nash-Williams] *For any k and graph, one can assign each edge a color from $\{1, \dots, k\}$ such that the resultant coloring does not have a monochromatic cycle iff for all subgraphs H it holds that*

$$k \cdot (\#vertices(H) - 1) \geq \#edges(H).$$

19.2 Matroid Axioms

Matroids was first defined as a link between vectors and graphs as a way of capturing the combinatorial essence of linear independence. They can be defined in many ways. Here is one. A finite **matroid** has a universe X and a nonempty collection $\mathcal{I}$ of subsets of X , known as the **independent sets**, that obey the following two axioms:

1. $\mathcal{I}$ is closed under subsets. Meaning, $K \in \mathcal{I}$, $J \subseteq K$ implies $J \in \mathcal{I}$.
2. If $J, K \in \mathcal{I}$ with $|K| > |J|$, then there is an $a \in K$ such that $J \cup \{a\} \in \mathcal{I}$.

These axioms are motivated by linear algebra. The first says that a subset of an independent set is independent, and the second is based on the idea that every independent set is extendable to a basis.

19.3 Matroid Examples

Here are some examples.

- sets of vectors. The two properties are well-known.
- sets: where every subset is considered independent.
- a **graphical matroid**: the universe is the set of edges of a graph and a collection of edges is defined to be independent if it does not contain a cycle. (Verifying the axioms is left as an exercise.)

Many of the concepts from linear independence carry over. For example, one can define a **base** (or basis) as a maximal independent set. And one can define a matroid in terms of a **rank function** (mirroring linear algebra where the rank is the size of the largest independent subset). The axioms for a rank function are:

1. It obeys $0 \leq \text{rank}(S) \leq |S|$.
2. If $S \subseteq T$ then $\text{rank}(S) \leq \text{rank}(T)$.
3. It is **submodular**: $\text{rank}(S \cup T) + \text{rank}(S \cap T) \leq \text{rank}(S) + \text{rank}(T)$

Because of the results where a trivial necessary condition turns out to be sufficient, one often checks for a “matroidal property” when investigating. Unfortunately such cases are rare. Matroids are now more studied for their own worth. There are special classes of matroids (e.g. binary matroids) and standard operations to build new matroids (e.g. deletion, minors, and duals).

19.4 Matroid Theorems

To prove our main matroid theorem we need the following lemma about refining the independent sets. We say that a set A **spans** B if $\text{rank}(A) = \text{rank}(A \cup B)$.

LEMMA. Consider a matroid with independent sets $\mathcal{I}$. For fixed set B , define $\mathcal{J}$ to be those sets in $\mathcal{I}$ that do not span B . Then $\mathcal{J}$ forms a matroid too.

PROOF. (1) *Closure under subsets*:

Assume $J \subset K \in \mathcal{J}$. Then K is independent and does not span B , which implies $K \cup \{a\}$ independent for some $a \in B$, which implies $J \cup \{a\}$ is independent. This implies J is independent and does not span B . Thus $J \in \mathcal{J}$.

(2) *Extensibility*:

Assume $J, K \in \mathcal{J}$ with $|K| > |J|$. Then both J and K are independent and neither spans B . The latter implies $K \cup \{a\}$ is independent for some $a \in B$. There are two cases.

Case 1: If $J \cup \{a\}$ is independent, then since $|J \cup \{a\}| < |K \cup \{a\}|$, there exists $k \in K$ such that $J \cup \{a, k\}$ is independent. But this implies that $J \cup \{k\}$ is independent and does not span B .

Case 2: If $J \cup \{a\}$ is dependent, then because J does not span B , we have $J \cup \{b\}$ independent for some $b \in B$. Again there exists $k \in K \cup \{a\}$ such that $J \cup \{b, k\}$ independent. This implies $k \neq a$, and so $J \cup \{k\}$ is independent and does not span B . $\square$

We next prove the independent representatives theorem:

THEOREM. [Rado] *For any matroid, a collection $\mathcal{C}$ of sets (from the universe) has a system of independent representatives iff for sub-collections $\mathcal{C}'$ it holds that*

$$\text{rank}(\bigcup \mathcal{C}') \geq |\mathcal{C}'|.$$

PROOF. We proceed by induction on the number k of sets in $\mathcal{C}$. The result is trivial if $k = 0$. Let $S_1, \dots, S_k$ be the sets in $\mathcal{C}$. Let $\mathcal{J}$ be the independent sets defined by reclassifying as dependent any (independent) set that spans S_k . We saw above that this forms a matroid, say M' .

Further, we claim the revised system obeys the hypothesis of the theorem. For, let $Z = \bigcup \mathcal{C}'$ where $\mathcal{C}'$ is a sub-collection of $S_1, \dots, S_{k-1}$. If Z does not span S_k , then its rank is unchanged. So,

$$\text{rank}'(Z) = \text{rank}(Z) \geq |\mathcal{C}'|.$$

If Z does span S_k , then originally $\text{rank}(Z) = \text{rank}(Z \cup S_k) \geq |\mathcal{C}'| + 1$. Since $\text{rank}(S_k) \geq 1$ (by hypothesis), there is a maximal independent set I in Z that contains exactly one element a of S_k . So $I - a$ is independent, does not span S_k , is contained in Z and has size $|\mathcal{C}'|$. That is,

$$\text{rank}'(Z) \geq \text{rank}(Z) - 1 \geq |\mathcal{C}'|.$$

Hence the revised system has a system T of representatives that are independent in M' and hence independent in the original matroid. But by the definition of independence in M' , T does not span S_k ; so one can add $a \in S_k$ to T while retaining independence (in the original matroid), which completes the inductive step. $\square$

The covering theorem:

THEOREM. [Edmonds & Fulkerson] *For any matroid and positive integer k , the universe X can be partitioned into k independent sets iff for all subsets A of X it holds that*

$$k \cdot \text{rank}(A) \geq |A|.$$

PROOF. Take k copies of the universe X and color each copy with a different color. Construct matroid $\mathcal{S}'$ with this universe, where a set is independent if its restriction to each color is an independent set. Then construct collection $\mathcal{C}$ with each set consisting of the k colored copies of one element of X . This collection obeys the conditions of the previous theorem and so there is a system T of independent representatives. Note that a system of representatives is an assignment of colors. Further, the system being independent means that T restricted to each color is independent. That is, we have partitioned X into k independent sets. $\square$